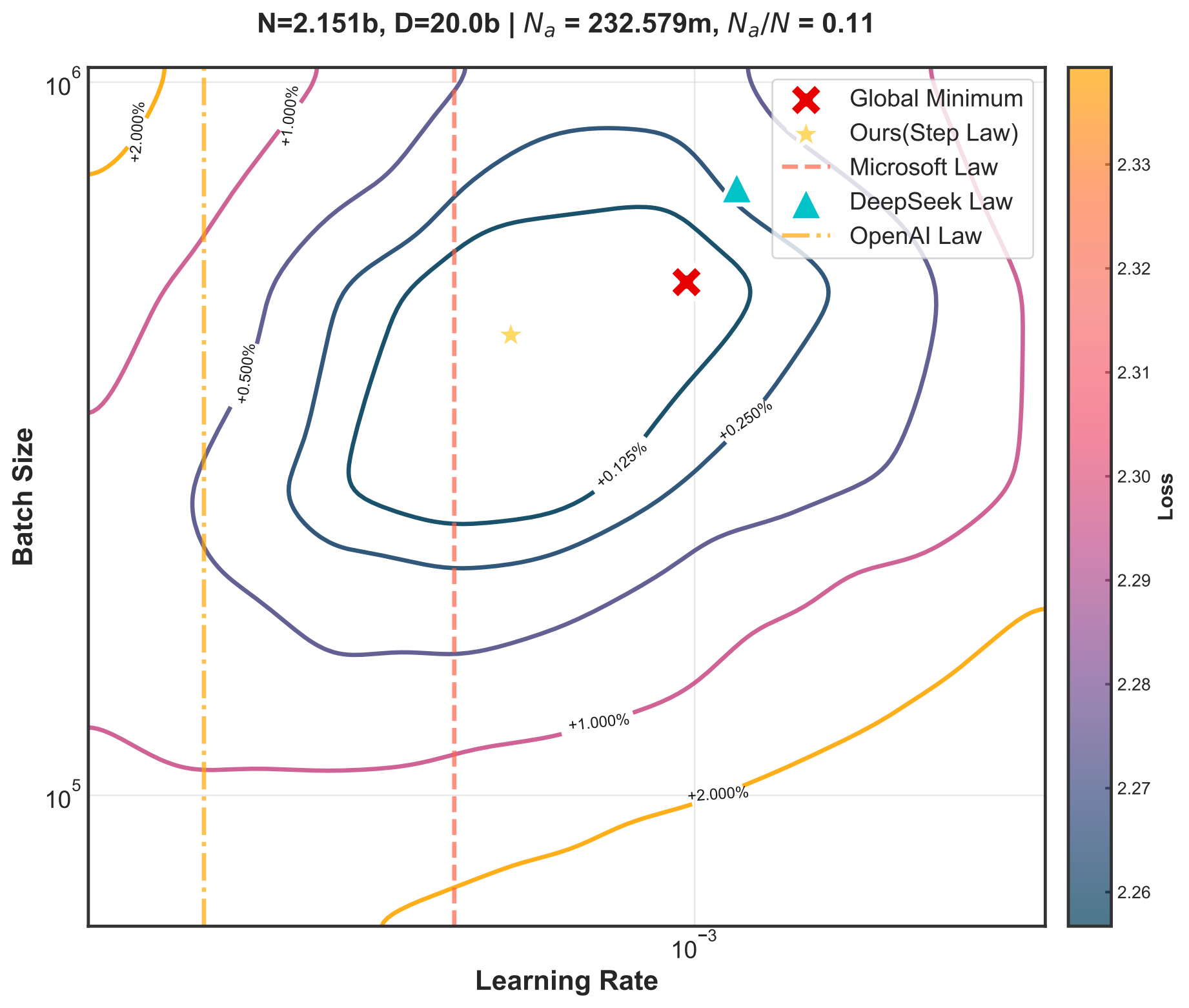
N=2.151b, D=20.0b | N_a = 232.579m, N_a/N = 0.11
Global Minimum
Ours(Step Law)
Microsoft Law
DeepSeek Law
OpenAI Law
+2.000%
+1.000%
+0.500%
+0.250%
+0.125%
+1.000%
+2.000%
Batch Size
Learning Rate
Loss
10^6
10^5
10^{-3}
2.33
2.32
2.31
2.30
2.29
2.28
2.27
2.26